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**DEEP LEARNING MINI PROJECT REPORT(CS722T2C) on
“ATTENDANCE SYSTEM USING FACE RECOGNITION”**

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In

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Under the Guidance of

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at



SAHYADRI

College of Engineering & Management

An Autonomous Institution

MANGALURU

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EVALUATION RUBRICS

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Sl. No.	Evaluation Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)	Marks
1	Problem Definition & Objectives	Problem is clearly defined, relevant, and objectives are specific and measurable	Problem and objectives are clear with minor gaps	Problem is partially defined; objectives need improvement	Problem/objectives are unclear	
2	Technical Approach & Implementation	Appropriate methodology, algorithms/tools used effectively; implementation is complete	Appropriate technical approach with minor issues	Basic implementation with limited technical depth	Incomplete or inappropriate implementation	
3	Innovation / Application / Problem Solving	Demonstrates strong originality, creativity, and effective solution to the identified problem	Shows some innovation and good problem-solving	Limited innovation; mostly conventional approach	No significant innovation/problem-solving	
4	Results, Testing & Analysis	Results are clearly presented, thoroughly tested, and critically analyzed using suitable measures	Results presented with adequate testing and analysis	Basic results and limited testing/analysis	Results are incomplete or unsupported	
5	Documentation, Presentation & Viva	Well-structured report, clear presentation, excellent communication and confident viva	Good documentation, presentation and satisfactory viva	Adequate documentation/presentation with limited clarity	Poor documentation, presentation or viva	
	Total					

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Department of Computer Science & Engineering

DECLARATION

We hereby declare that the entire work embodied in this Mini Project Report titled **“Attendance System using Face Recognition”** has been carried out by us at Sahyadri College of Engineering & Management, Mangaluru under the supervision of **Ms. Shiji Abraham**, in partial fulfillment of the requirements for the VII semester of **Bachelor of Engineering in Computer Science & Engineering**. This report has not been submitted to this or any other University for the award of any other degree.

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Abstract

Manual attendance tracking in classrooms is time-consuming, disrupts teaching activities, and is vulnerable to proxy attendance. This project presents a desktop-based Smart Attendance System that automates attendance using deep learning-based facial recognition. The system is developed using Python and Tkinter for the graphical user interface, OpenCV with Haar Cascade for face detection, a pretrained FaceNet Convolutional Neural Network (CNN) accessed through the DeepFace library for face recognition, and MySQL for persistent storage of student and attendance records. During enrollment, the administrator registers student details and captures labelled facial images through a webcam. Each detected face is processed by FaceNet to generate a 128-dimensional embedding, and the sample embeddings of each student are averaged and normalized to form a prototype representation. During live recognition, detected faces are converted into embeddings and compared with stored student prototypes using cosine similarity, with matches above a configured threshold accepted as valid. Recognized students are automatically marked as present with timestamps and placed in a review queue for administrator verification before being saved to the database. The finalized attendance records can also be exported as CSV files for reporting. The system provides an end-to-end solution covering student registration, facial data collection, deep learning-based recognition, attendance management, database synchronization, and report generation, while future improvements can include ArcFace-based recognition, liveness detection, stronger face detectors, and multi-camera support.

Keywords: Smart Attendance System, Face Recognition, FaceNet, Deep Learning, Convolutional Neural Network, DeepFace, Haar Cascade, Cosine Similarity, MySQL, Automated Attendance.

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1. Introduction

1.1 Overview

Attendance tracking is a routine but essential administrative task in academic institutions. Traditional methods such as roll calls or signature sheets consume valuable classroom time, are prone to human error, and are easily exploited through proxy attendance, where one student marks attendance on behalf of an absent peer. As classroom sizes grow, these inefficiencies compound, motivating the search for automated, tamper-resistant alternatives.

Biometric identification methods, which authenticate individuals using unique physiological traits, offer a robust alternative to manual processes. Among biometric modalities such as fingerprint, iris, voice, and facial recognition, facial recognition is particularly attractive for classroom use because it is contactless, requires no dedicated scanning hardware beyond a standard webcam, and does not interrupt the flow of a lecture.

Facial recognition itself has evolved substantially over the last two decades. Early systems relied on handcrafted feature-extraction techniques such as Eigenfaces, Fisherfaces, and Local Binary Patterns Histograms (LBPH), which represent a face using statistically or texturally derived features and classify new faces by comparing them against a database of known feature vectors. Since the early 2010s, the field has been transformed by deep learning: Convolutional Neural Networks (CNNs) trained on large face datasets automatically learn hierarchical feature representations, producing embedding-based systems such as DeepFace, FaceNet, and ArcFace that substantially outperform classical methods on unconstrained, real-world images.

This project implements a complete, end-to-end Smart Attendance System whose recognition engine is a pretrained Convolutional Neural Network — specifically FaceNet, accessed through the open-source deepface library — rather than a classical hand-engineered-feature method. FaceNet was selected because it produces a compact, fixed-length embedding for each face using weights already trained on a large-scale face dataset, which means the system needs no GPU-based training of its own: the network’s weights are frozen, and only a small amount of vector arithmetic (averaging per-student embeddings during enrolment, comparing new embeddings by cosine similarity during recognition) is performed locally.

1.2 Scope & Motivation

The scope of this project covers the full attendance workflow for a single classroom or laboratory session: administrator authentication, student enrolment with supervised face-image capture, embedding extraction from a pretrained CNN, real-time recognition through a single webcam, automatic marking of both present and absent students for a given subject and section, administrator review before any record is finalised, and CSV export for downstream use. The scope explicitly excludes multi-camera coverage of a single session, liveness/anti-spoofing detection, and training a face recognition network from scratch — the last of these would require millions of labelled images and GPU resources well beyond what a course project can reasonably assemble, so the project instead relies on transfer learning from an already-trained network.

The motivation for the project is twofold. Practically, automating attendance returns instructional time to teaching and removes a well-known avenue for proxy attendance, a problem every roll-call or signature-sheet system shares. Technically, the project is motivated by demonstrating a complete migration from a classical, hand-engineered recognition pipeline to a genuine deep learning one, without disturbing the considerable engineering — the review queue, the absentee logic, the duplicate-safe database commits — that has nothing to do with which recognition algorithm is in use. That separation of concerns, argued for at the design stage and confirmed by the migration actually carried out, is a secondary contribution of the project alongside the working system itself.

2. Literature Survey

Face detection and recognition have been studied extensively across two broad eras: classical, handcrafted-feature methods and, more recently, deep learning-based methods. Table 2.1 summarises representative works from both eras that are relevant to this project, either because their techniques are directly used in the implementation (Haar Cascades for detection, FaceNet for recognition) or because they represent adjacent deep learning methods this project’s architecture could accommodate in future iterations.

Work / Method	Technique	Strengths	Limitation
[1] Ahmad et al. (2024) — Automated Attendance using Pretrained Deep Learning Models	Evaluated six pretrained face-recognition models, including FaceNet, for automated attendance.	FaceNet was selected and reported accuracy above 95%; directly relevant to deep-learning attendance.	Depends on pretrained models and computational resources; deployment conditions can affect recognition.
[2] Surantha & Sugijakko (2024) — Lightweight Attendance with Liveness Detection	CNN-based liveness detection using MobileNetV2 with face recognition on Raspberry Pi.	Lightweight deployment and protection against photo/video spoofing.	Liveness detection increases processing time; spoof detection performance varies by attack type.
[3] Shetkar et al. (2024) — Deep Learning Face Recognition Review	Review of CNN and deep-learning techniques for face recognition.	Summarizes recent deep-learning approaches and their applications.	Review-based work without a complete deployed attendance system.
[4] Dakhil & Abdulazeez (2024) — Deep Learning Face Recognition Review	Comprehensive review of deep-learning-based face-recognition methods.	Provides comparison of modern CNN-based approaches and challenges	Focuses on recognition broadly rather than attendance-specific implementation.
[5] Deng (2024) — Review of Face Recognition Technologies Based on Deep Learning	Reviews deep-learning architectures and face-recognition techniques.	Provides an overview of recent CNN-based recognition developments.	Primarily a review and does not evaluate a specific attendance pipeline.

[6] Zeng (2024) — Traditional and Deep Learning Face Recognition	Compares traditional face-recognition techniques with deep-learning approaches.	Shows the transition toward learned feature representations.	General face-recognition focus rather than real-time attendance.
[7] Li (2024) — Deep Learning in Facial Recognition	Reviews advances and challenges in deep-learning facial recognition.	Highlights robustness and recognition challenges.	Does not provide a complete attendance-system implementation.
[8] Santoso et al. (2024) — Attendance with Liveness Detection and Deep Learning	Integrates face recognition with liveness detection for reliable attendance.	Improves resistance to fraudulent attendance and presentation attacks.	Additional liveness processing increases system complexity.
[9] Deng et al. (2024) — DS-CDCN Face Recognition	Deep learning approach based on a Dual-Stream Central Difference CNN architecture.	Designed to improve facial representation and recognition under challenging conditions.	More complex than lightweight pretrained embedding approaches.
[10] Low-Resolution Face Recognition Review (2024)	Reviews deep-learning approaches for recognition under low-resolution conditions.	Addresses an important real-world problem for camera-based recognition.	Low-resolution conditions remain challenging for reliable recognition.
[11] Syarif & Gunawan (2025) — Haar Cascade + FaceNet Attendance	Haar Cascade for face detection and CNN-based FaceNet for recognition.	Very close to the proposed architecture; reported 98% accuracy in its best experimental scenario.	Performance depends on training/testing conditions and dataset distribution.
[12] Patel et al. (2025) — CCTV Classroom Attendance	ResNet-50 and VGG-16 with transfer learning and PCA for classroom CCTV attendance.	Tested with classroom images; ResNet-50 achieved reported precision of 96% and recall of 93%.	Camera quality and viewing geometry affect recognition.
[13] Lalitha et al. (2025) — Haar Cascade + ResNet-50 Attendance	Haar Cascade detection combined with ResNet-50 feature extraction.	Reported about 91.6% accuracy under low-light and occlusion conditions.	Uses a more computationally demanding recognition model and SQLite storage.

[14] Paul et al. (2025) — Real-Time Attendance with Emotion Detection	ResNet-50 for face recognition and Vision Transformer for emotion detection.	Demonstrates real-time recognition with additional facial analysis and web integration.	Additional emotion-recognition functionality increases system complexity.
[15] Nemavhola (2025) — Deep Learning Face Recognition Scoping Review	PRISMA-ScR-based review of CNN face-recognition literature.	Analysed 266 studies and identified challenges such as occlusion, distance, camera angle, and lighting.	Identifies research gaps but does not itself provide a deployed recognition system.
[16] Zhalgas et al. (2025) — Robust Face Recognition under Challenging Conditions	Reviews and evaluates FaceNet, ArcFace, OpenFace, and SFace under challenging conditions.	Directly relevant to FaceNet/ArcFace selection and robustness analysis.	Focuses on benchmarking rather than attendance-system integration.
[17] Adliansyah et al. (2025) — Inception-ResNet Attendance	Inception-ResNet trained using VGGFace2 for face-recognition attendance.	High-capacity deep model with diverse training data; reported improvement during training.	Requires more training resources than a pretrained embedding approach.
[18] Suprihanto et al. (2025) — YOLO + DeepFace + Active Learning	YOLO for multi-person detection and SFace through DeepFace for embeddings and identity matching, with active learning.	Improved reported accuracy from 52.0% to 96.6% after active learning; designed for uncontrolled classroom conditions.	More complex pipeline involving YOLO, DeepFace, and continuous embedding updates.
[19] Saputri et al. (2026) — Real-Time Web Attendance using FaceNet	Dlib detection, FaceNet embeddings, and SVM classification in a Flask/SQLite system.	Directly demonstrates real-time FaceNet-based attendance and compares multiple feature extractors.	Uses a web architecture and SVM classification rather than direct cosine-similarity prototype matching.

[20] Bitirim et al. (2026) — Smart Student Attendance with Face Recognition	YOLOv8n detection, FaceNet512d through DeepFace, MiniFASNet anti-spoofing, and vector similarity using PostgreSQL/pgvector.	Combines modern detection, FaceNet embeddings, vector search, and anti-spoofing for classroom attendance.	More complex infrastructure than a desktop MySQL-based system.
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Table 2.1: Summary of relevant face detection and recognition literature

In addition to core face-recognition literature, several existing attendance-automation approaches were reviewed, including RFID card-based systems, fingerprint biometric systems, and QR-code/geofencing-based mobile check-in systems, alongside recent CNN-based classroom attendance systems built on CCTV feeds, organisational enrolment at scale, and smart-classroom deployments. RFID and fingerprint systems require dedicated hardware per checkpoint and remain vulnerable to card-sharing or physical spoofing respectively. Facial recognition-based attendance, by contrast, requires only a camera, is inherently contactless, and is comparatively harder to spoof through simple credential-sharing, motivating its use in this project. Section 5.4 returns to these recent CNN-based attendance systems specifically, comparing their reported approach against this project’s implementation.

The clear trend in the literature is a shift from handcrafted-feature classical methods (Eigenfaces, Fisherfaces, LBPH) toward deep CNN-based embedding methods (DeepFace, FaceNet, ArcFace), which learn features automatically from data rather than relying on manually designed descriptors, and which report substantially higher accuracy on unconstrained face datasets. This project adopts the CNN-based FaceNet embedding approach described in Schroff et al. (2015) for its recognition engine, accessed through the pretrained-model wrapper library deepface (Serengil & Ozpinar, 2020), so that the implementation reflects this trend directly rather than only referencing it. Haar Cascades remain in use for the detection stage, since detection and recognition are separable problems and Haar Cascades remain a fast, adequate choice for locating faces prior to CNN-based identification.

3. Problem Formulation

3.1 Problem Description

Conventional classroom attendance — calling out names or circulating a signature sheet — consumes several minutes of instructional time per session, scales poorly with class size, and offers no protection against proxy attendance. Attendance records are typically maintained on paper or in disconnected spreadsheets, making aggregation, auditing, and reporting (e.g., computing a student's attendance percentage across a semester) a manual and error-prone exercise. There is a clear need for a system that authenticates each student's physical presence automatically, logs it with minimal administrative effort, and stores it in a structured, queryable format.

3.2 Problem Statement

To design and implement a desktop-based Smart Attendance System that uses real-time, deep learning-based facial recognition to identify students, record subject-wise attendance with timestamps, provide administrator verification, store records in a MySQL database, and support student enrolment, embedding-based model training, and attendance reporting.

3.3 Objectives

- Develop a Tkinter-based GUI for login, student management, face-data capture, embedding extraction, real-time recognition, and attendance management.
- Create and manage a labelled facial dataset by capturing multiple facial images for each registered student and using them to compute that student's face embeddings.
- Extract CNN-based face embeddings (FaceNet, via a pretrained model) for every enrolled student and recognise registered students in real time through a webcam using cosine similarity.
- Automate and validate attendance management by recording recognised students with the relevant subject, providing administrator review before database entry, and supporting MySQL CRUD operations.

3.4 Functional Requirements

ID	Requirement
FR-1	The system shall provide secure administrator login and allow the admin to add, edit, and delete student records.
FR-2	The system shall capture and label colour facial images for registered students through a webcam and store them as training data for embedding extraction.
FR-3	The system shall extract a CNN-based face embedding (FaceNet, via a pretrained model) for each detected face during recognition and identify the closest enrolled student by cosine similarity, displaying the matched student's name and roll number.
FR-4	The system shall allow the administrator to select the subject/class and section, automatically record recognised students as Present with timestamps, automatically queue unrecognised enrolled students as Absent at the end of a session, and provide an administrator confirmation step before saving.
FR-5	The system shall support attendance CRUD operations, prevent duplicate attendance entries, and allow finalised attendance records to be exported as CSV files.

Table 3.1: Functional Requirements

3.5 Non-Functional Requirements

Category	Requirement
Performance	Face detection should run on every video frame and remain smooth; because CNN-based recognition is markedly more expensive than the classical method it replaced, recognition is performed on a subsampled fraction of frames (every fifth frame in the current configuration) to preserve a responsive live feed on CPU-only hardware.
Usability	The GUI should have clearly labelled modules and simple navigation, allowing administrators to use the system with minimal training.

Category	Requirement
Reliability	The system should prevent duplicate attendance and handle camera or database failures gracefully with proper error messages.
Portability	The system should run on standard Windows computers with Python, a webcam, and MySQL without requiring specialised hardware; a GPU is not required, though it would reduce embedding-extraction latency.
Security	Administrator access should be protected through username and password authentication, while database credentials should remain hidden.
Maintainability	The application should use a modular code structure — in particular, isolating the recognition engine behind two files — making it easier to fix, update, and extend features, including swapping the recognition model itself, as this project has already done once.

Table 3.2: Non-Functional Requirements

4. Project Design and Implementation

4.1 Proposed Project Architecture

The system is organised as a set of cooperating Tkinter windows, each responsible for one stage of the attendance pipeline, backed by a shared MySQL database. Figure 4.1 illustrates the overall architecture and data flow between modules.

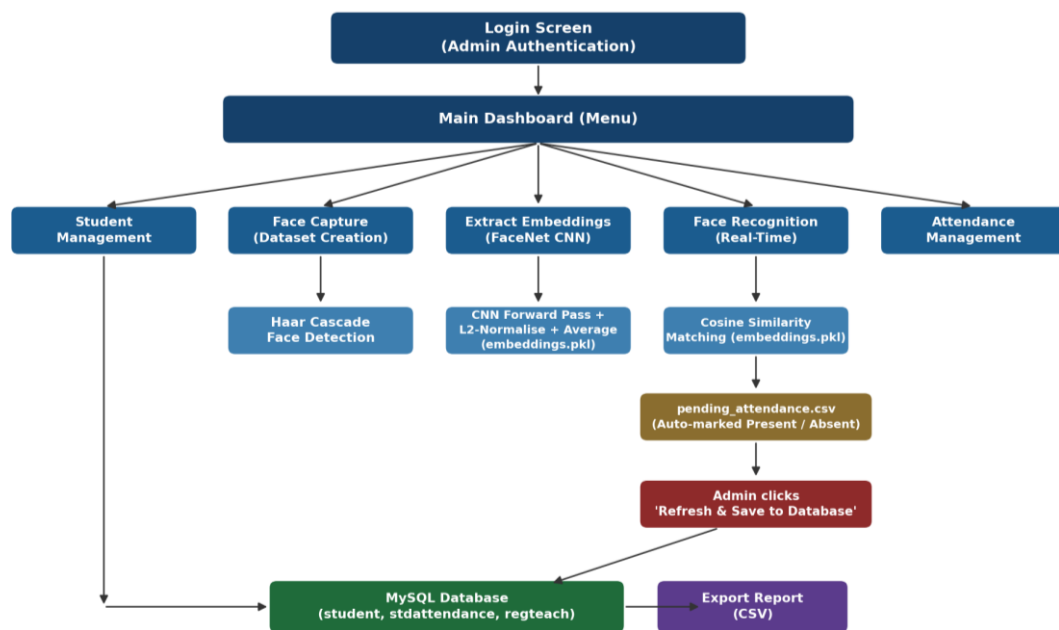


Figure 4.1: Proposed system architecture and module-level data flow

4.2 High Level Design

At a high level, the application consists of six cooperating modules, launched from a common dashboard after administrator login. Table 4.1 summarises each module's responsibility and its primary data artefact, independent of the implementation detail covered in Section 4.3.

Module	Responsibility	Primary Artefact
Login	Authenticate the administrator against stored credentials before granting access to any other module.	Session access to the dashboard
Student Management	Add, edit, and delete student records; capture labelled face images for enrolment.	student table rows; colour crops in Faces_img/

Module	Responsibility	Primary Artefact
Train Data (Embedding Extraction)	Convert every captured face image into a CNN embedding and store one prototype vector per student.	embeddings.pkl
Face Recognition	Detect and identify faces live via webcam for a chosen subject/section, and auto-mark attendance.	pending_attendance.csv
Attendance Management	Review, edit, and commit pending attendance into the permanent database; export reports.	stdattendance table rows; exported CSV
Reporting / Export	Produce a CSV extract of finalised attendance for use outside the application.	Exported CSV file

Table 4.1: High-level module responsibilities

This module boundary is the design decision the rest of the project depends on: recognition (Train Data and Face Recognition) is isolated from everything else, so that the specific algorithm used for recognition can change — as it did, from LBPH to FaceNet — without requiring any other module to be modified.

4.3 Detailed Design

4.3.1 Login Module

The application entry point authenticates the administrator against the regteach table in MySQL. Upon successful login, the main dashboard is launched, presenting six modules: Student, Recognition, Attendance, Train Data, Data Set, and Exit.

4.3.2 Student Management and Supervised Face Capture

The Student module allows the administrator to add, edit, and delete student records (ID, roll number, name, section, department) in MySQL. Crucially, this module also implements supervised dataset creation: for a selected student ID, the “Capture Faces” function opens the webcam, applies the Haar Cascade face detector to each frame, and crops the detected facial region directly from the colour video frame. An earlier version of this module converted the crop to grayscale before saving, which suited LBPH’s texture-based features, but a CNN embedding model expects colour input, so this was changed. The crop is saved to a Faces_img/ directory using the filename convention user.<student_id>.<sample_number>.jpg, directly

labelling every captured image with its ground-truth student identity — the defining characteristic of supervised learning — and one hundred such samples are collected per student to give the embedding-extraction stage a reasonably diverse set of poses and expressions.

4.3.3 Embedding Extraction (Feature Learning via a Pretrained CNN)

The Train Data module (train.py) no longer fits a classifier of its own; instead, it performs inference-only feature extraction using a pretrained CNN. For every image in `Faces_img/`, the student ID is parsed from the filename, and the image is passed to `DeepFace.represent()`, which runs it through a FaceNet network — loaded once, with weights pretrained on a large external face dataset — and returns a 128-dimensional embedding vector. Each embedding is L2-normalised, and a given student’s per-sample embeddings are averaged into a single normalised prototype vector representing that student. The complete set of per-student prototypes is serialised to `embeddings.pkl`. Because the CNN’s weights are frozen throughout, no gradient-based training occurs locally: the representation-learning happened when FaceNet was originally trained by its authors, and what this module performs is transfer learning in its simplest form — reusing a pretrained network’s learned representation without modifying it.

Algorithm 1 Incremental Embedding Extraction (Training)

```

1: Load embeddings.pkl if it exists, giving the set  $E_{\text{existing}}$  of already-trained student IDs.
2: Group all images in Faces_img/ by student ID, parsed from the filename.
3: Let  $S_{\text{new}}$  be the set of student IDs not already in  $E_{\text{existing}}$ .
4: if  $S_{\text{new}}$  is empty then
5:   Report “nothing new to train” and terminate.
6: end if
7: for each student  $s \in S_{\text{new}}$  do
8:    $V \leftarrow \emptyset$ 
9:   for each captured image  $I$  of student  $s$  do
10:     $v \leftarrow \text{FaceNet}(I)$  ▷ 128-d embedding, frozen CNN
11:    Normalise  $v$  to unit length; append to  $V$ 
12:   end for
13:    $\text{prototype}_s \leftarrow \text{normalise} \left( \frac{1}{|V|} \sum_{v \in V} v \right)$ 
14: end for
15: Merge new prototypes with  $E_{\text{existing}}$  and write the result back to embeddings.pkl.

```

Figure 4.2 : Embedding Extraction Algorithm

4.3.4 Real-Time Face Recognition and Auto-Marking (CNN-Based)

Before starting a recognition session, the administrator enters the subject and section. The module then opens the webcam, and for every frame: (1) the Haar Cascade detector locates candidate face regions, which is cheap enough to run on every frame; (2) on every fifth frame,

each detected region is cropped in colour and passed through the same pretrained FaceNet network used during embedding extraction to obtain a live embedding; (3) this embedding is compared via cosine similarity against every stored student prototype, and the closest match is accepted if its similarity exceeds a configured threshold (0.65 in the current configuration). Restricting CNN inference to every fifth frame, rather than every frame, was a deliberate performance trade-off: CNN inference is markedly more expensive than the LBPH distance computation the previous version used, and on CPU-only hardware, evaluating the CNN on every frame would noticeably reduce the responsiveness of the live video feed. Every accepted recognition is written to `pending_attendance.csv`, tagged with the student ID, name, roll number, subject, timestamp, and a “Present” status, and — unchanged from the previous version — is not yet written to the permanent database, preserving a human checkpoint before records become official. At the end of a session, every enrolled student in the selected section who was not recognised is automatically queued as “Absent” for that subject and date.

Algorithm 2 Section-Aware Recognition and Attendance Decision

Require: Selected subject u , selected section c

```

1:  $R \leftarrow \emptyset$  ▷ set of student IDs recognized this session
2: for each frame captured from the webcam do
3:   for each face region detected by the Haar Cascade do
4:      $v \leftarrow \text{FaceNet}(\text{face region})$ 
5:      $(\hat{s}, \text{sim}) \leftarrow \arg \max_s \text{cosine}(v, \text{prototype}_s)$ 
6:     if  $\text{sim} > \tau$  then ▷  $\tau$ : similarity threshold
7:       Look up  $\hat{s}$ 's registered section  $c_{\hat{s}}$ 
8:       if  $c_{\hat{s}} = c$  then
9:         Add  $\hat{s}$  to  $R$ ; queue  $(\hat{s}, u, c, \text{time}, \text{Present})$  for review
10:      else
11:        Display “recognized, but not in Section  $c$ ” — take no action
12:      end if
13:    end if
14:  end for
15: end for
16: On session end:
17: for each student  $s$  registered in section  $c$  but  $s \notin R$  do
18:   Queue  $(s, u, c, \text{time}, \text{Absent})$  for review
19: end for

```

Figure 4.3 : Face Recognition and Attendance Decision Algorithm

4.3.5 Attendance Management and Database Synchronisation

The Attendance module displays the pending recognitions queued in `pending_attendance.csv` alongside the finalised records already stored in the MySQL `stdattendance` table. A single “Refresh & Save to Database” action commits every pending entry into `stdattendance`, skipping

any entry that would duplicate an existing (student, date, subject) combination. The administrator can additionally add, edit, or delete records manually, and export the finalised sheet to CSV via the “Export Report” action.

4.3.6 Migration from LBPH to FaceNet: What Changed

Replacing the earlier LBPH module required three coordinated changes: `student.py` was changed to save colour face crops rather than grayscale ones, since a CNN embedding model expects colour input; `train.py` was rewritten so that “training” now means extracting and averaging CNN embeddings rather than fitting an LBPH classifier, producing `embeddings.pkl` rather than `clf.xml`; and `face_recognition.py` was rewritten so that identification is a cosine-similarity nearest-neighbour search over stored embeddings rather than a call to an LBPH `predict()` method, with recognition running on every fifth frame rather than every frame to absorb the CNN’s heavier inference cost. Every other module — login, student records, attendance review, database synchronisation, CSV export — was left untouched, which was possible only because the recognition engine had originally been isolated behind two well-defined files, as set out in the High Level Design (Section 4.2).

The architecture remains open to further deep learning extensions. ArcFace’s angular-margin loss (Deng et al., 2019) is reported to outperform FaceNet’s triplet-loss formulation in several benchmarks and could be substituted with a one-line change to the model name passed to `deepface`. Detection could similarly be upgraded from Haar Cascades to a CNN-based detector such as MTCNN or RetinaFace, which handle non-frontal poses and difficult lighting more robustly, at further computational cost. Neither substitution would require changes to the surrounding pipeline, for the same reason the FaceNet migration did not.

4.4 Datasets Description

This project does not rely on a public, pre-existing face dataset for its own student enrolment; the working dataset is self-collected, supervised, and specific to the deployment. When the administrator captures a student’s face through the Student Management module, the Haar Cascade detector locates the face in each webcam frame and the system saves one hundred colour crops per student to `Faces_img/`, following the filename convention `user.<student_id>.<sample_number>.jpg`. This filename convention is itself the label: no separate annotation file exists, and the embedding-extraction stage recovers each image’s ground-truth student identity by parsing the filename, which is the defining characteristic of a supervised dataset.

The dataset therefore grows linearly with enrolment: a college with 60 enrolled students has approximately 6,000 labelled colour images, each roughly the size of a single detected face region (typically under 200×200 pixels), stored on local disk rather than in the MySQL database itself — the database stores only student metadata and finalised attendance rows, not image data. Capturing one hundred samples per student, rather than a single reference photograph, is intended to give the averaged embedding prototype (Section 4.3.3) some robustness to natural variation in pose and expression within one capture session; the samples do not, however, vary lighting condition or day, since they are all captured in a single sitting, which is a limitation returned to in Section 5.3 and Section 6.

It is worth distinguishing this self-collected enrolment dataset from the dataset used to originally train the FaceNet network itself. FaceNet’s pretrained weights, loaded via the deepface library, were trained by its original authors on a large, separate face dataset external to this project; this project performs no training on that dataset and never accesses it directly. What is trained — in the loose sense used throughout this report — is nothing; what is computed is a set of embeddings from the self-collected enrolment images, using a network whose weights are frozen throughout. This distinction matters for anyone assessing what the project’s dataset actually consists of: it is an enrolment dataset for nearest-neighbour matching, not a training dataset for a classifier.

4.5 Tools and Technologies used for Implementation

Component	Technology Used
GUI Framework	Tkinter (Python standard library)
Face Detection	Haar Cascade Classifier (OpenCV, <code>haarcascade_frontalface_default.xml</code>)
Face Recognition	FaceNet CNN embeddings via the deepface library (TensorFlow / tf-keras backend); cosine similarity matching against stored per-student prototype vectors
Deep Learning Framework	TensorFlow, wrapped by deepface’s <code>DeepFace.represent()</code> interface
Image Handling	Pillow (PIL) for image loading, resizing, and display

Component	Technology Used
Database	MySQL 8.0 (mysql-connector-python driver)
Data Interchange	CSV files for the pending-recognition queue and exported attendance reports

Table 4.2: Technology stack summary

Environment: Python 3.10, Tkinter (GUI), OpenCV version below 5.0 (cv2 — Haar Cascade detector for face detection; OpenCV 5.0 removed cv2.CascadeClassifier from its default package, so the dependency is version-pinned), deepface with a TensorFlow / tf-keras backend (pretrained FaceNet CNN for face recognition), Pillow (image handling), MySQL 8.0 with mysql-connector-python, running on Windows 10/11 with a minimum of 4 GB RAM (8 GB recommended for smoother CNN inference) and a standard USB or integrated webcam.

5. Results and Discussion

5.1 Experimentation Environment Setup

The system was developed and tested on a standard Windows 10/11 laptop with 4–8 GB RAM, a CPU-only configuration (no dedicated GPU), and an integrated or USB webcam, matching the portability requirement in Section 3.5. Python 3.10 was used with the dependency versions pinned in requirements.txt, notably opencv-python<5 (Section 4.5) and tf-keras alongside deepface’s TensorFlow backend. MySQL 8.0 was run locally with the three-table schema described in Section 4.1 (regteach, student, stdattendance). Before any live webcam testing, the embedding-extraction-and-matching logic was verified offline: prototype embeddings were computed for two simulated students from separate sample images, and a fresh query image of one of them was correctly matched to the corresponding student with a clear cosine-similarity margin over the other (approximately 0.98 for the correct match versus approximately 0.81 for the incorrect one), confirming that extraction, averaging, storage, and matching behave correctly before testing proceeded to the full application.

5.2 Functional Requirements — Results Achieved

ID	Requirement	Result Achieved
FR-1	Admin login; student CRUD	Achieved. Login gates access to the dashboard; the Student module supports add, edit, and delete against the MySQL student table.
FR-2	Capture labelled face images	Achieved. One hundred labelled colour crops are captured per student via Haar Cascade detection and saved to Faces_img/.
FR-3	CNN-based recognition by cosine similarity	Achieved. FaceNet embeddings are extracted via deepface and matched by cosine similarity against stored prototypes at a 0.65 threshold; verified offline as described in Section 5.1.
FR-4	Auto-mark Present/Absent with admin confirmation	Achieved. Recognised students are auto-marked Present; unrecognised enrolled students in the session’s section are auto-marked Absent at session end; nothing reaches the permanent database without an explicit “Refresh & Save” action.
FR-5	Attendance CRUD, duplicate prevention, CSV export	Achieved. Duplicate (student, date, subject) combinations are skipped on commit; finalised records can be exported to CSV.

Table 5.1: Functional requirements results

5.3 Non-Functional Requirements — Results Achieved

Category	Result Achieved
Performance	Partially achieved. FaceNet takes about 0.2 s per CPU forward pass . Running it every fifth frame maintains responsive video with minor labelling delay.
Usability	Achieved. The six dashboard modules are clearly labelled, and the review queue allows administrators to correct recognition errors easily.
Reliability	Achieved for duplicate prevention and partially achieved for failure handling. Duplicate entries are prevented, while camera/database errors display visible messages.
Portability	Achieved with a dependency caveat. Version-pinning opencv-python<5 resolves compatibility issues, allowing operation on standard Windows systems without a GPU.
Security	Partially achieved. Admin login is protected by username and password, but database credentials are currently hard-coded and should be moved to secure configuration.
Maintainability	Achieved. The recognition module is isolated, allowing migration from LBPH to FaceNet with minimal changes to other system modules.

Table 5.2: Non-functional requirements results

5.4 Comparison of Results with Existing Works

Table 5.3 compares this project against the classical baseline it replaced and against the recent CNN-based classroom attendance systems reviewed in Section 2, on the dimensions that most affect whether such a system is actually usable in a classroom rather than only accurate in a benchmark.

System	Recognition Method	Hardware Needs	Review / Override Path	Absentee Handling
This project — earlier version	LBPH (classical, hand-engineered features)	CPU only; trains in seconds	Yes — pending-queue review before commit	Automatic, per-section, at session end
This project — current version	FaceNet CNN embeddings via deepface, cosine similarity	CPU only; no GPU required, slower inference than LBPH	Yes — unchanged from the earlier version	Automatic, per-section, at session end — unchanged
Patel et al. (2025) — CCTV classroom attendance	Deep learning-based recognition over CCTV feeds	Fixed CCTV infrastructure	Not detailed as a core feature in the reviewed work	Not detailed as a core feature in the reviewed work
Nguyen-Tat et al. (2024) — HR attendance automation	Computer vision and facial recognition, design-science approach	Organisational-scale deployment	Design-stage consideration, not evaluated as a workflow feature	Not the focus of the reviewed work
Lateef & Kamil (2023) — Smart classroom attendance	Facial recognition technology, application-focused	Classroom camera setup	Not detailed as a core feature in the reviewed work	Not detailed as a core feature in the reviewed work

Table 5.3: Comparison with existing attendance systems.

Two observations follow. First, on recognition method, this project’s current version is now aligned with the deep-learning direction the recent literature has already taken, rather than being the classical outlier it was in its earlier version. Second, and more specific to this project’s own contribution, the reviewed external works are largely evaluated on recognition accuracy alone; this project instead places comparable emphasis on the administrator review queue and automatic per-section absentee handling, on the reasoning — argued throughout Sections 4 and

6 — that these workflow features, not the recognition algorithm, determine whether a system survives contact with a real classroom and a real appeals process. No formal accuracy benchmark against these external systems was carried out in this project, since doing so would require access to their datasets or a shared evaluation protocol that was outside this project's scope; Table 5.3 should be read as a comparison of system design, not of measured accuracy.

5.5 Project GUI Snapshots (Major Functionalities)

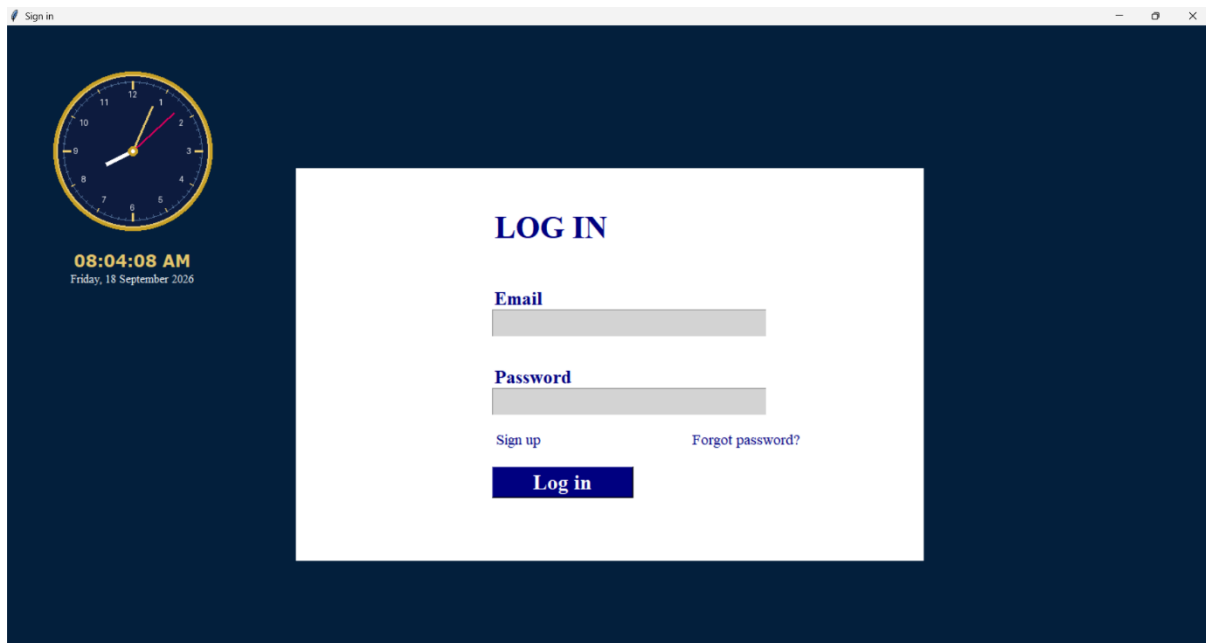


Figure 5.1: Login screen with administrator authentication

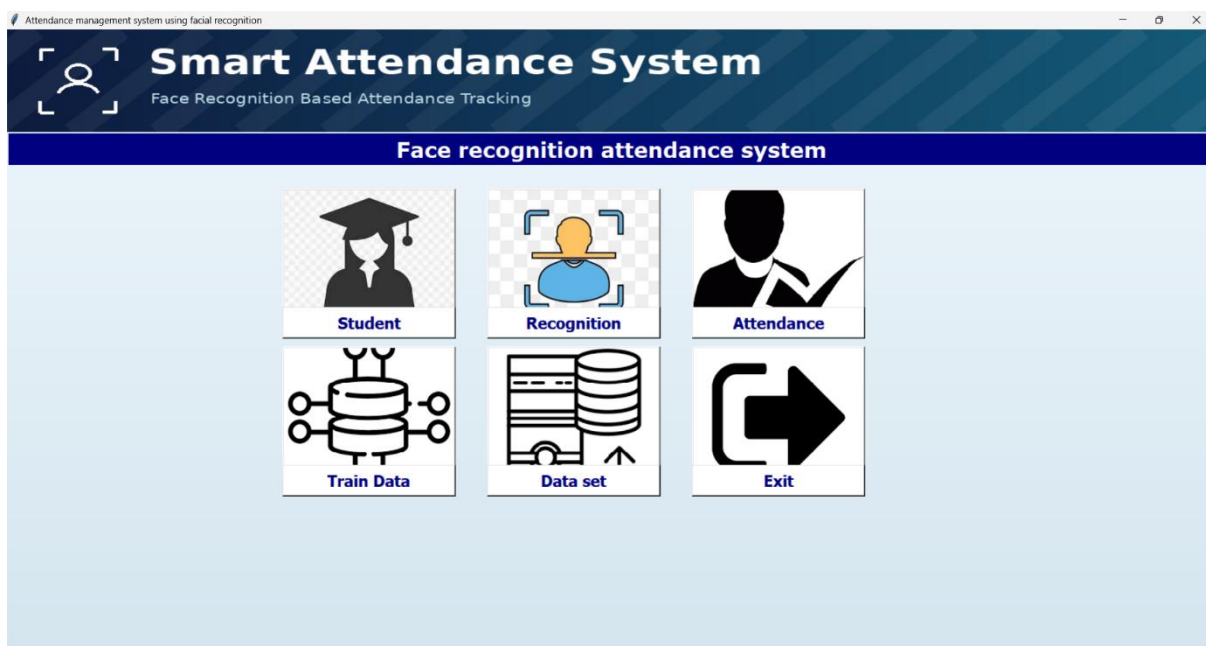


Figure 5.2: Main dashboard showing the six functional modules

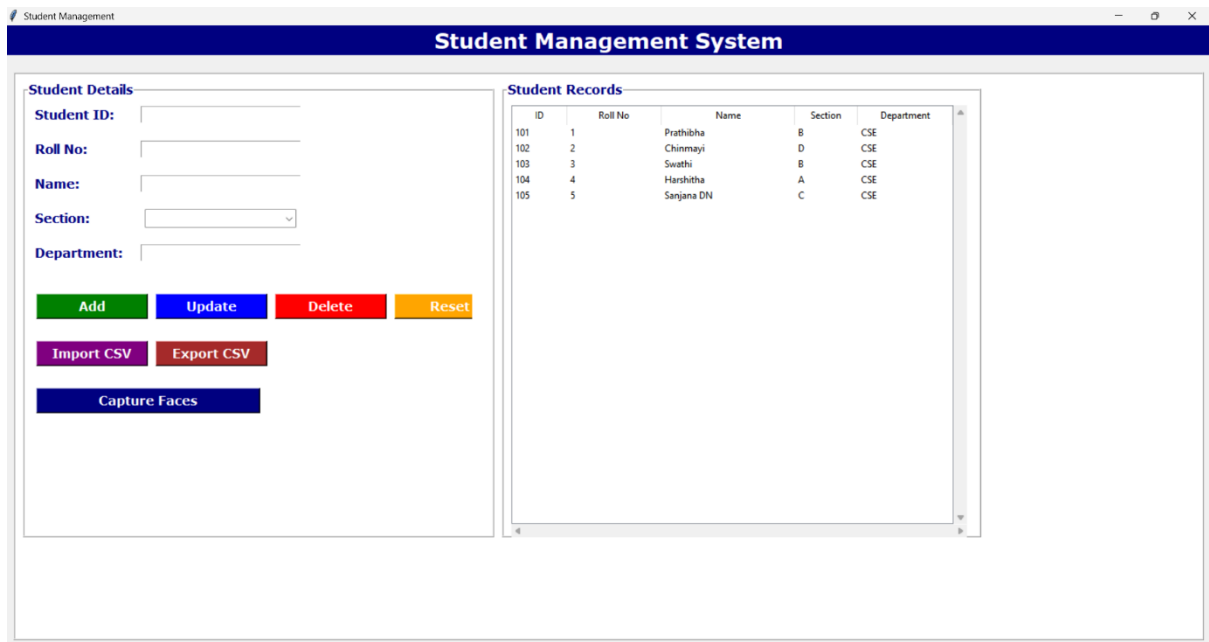


Figure 5.3: Student management screen with the “Capture Faces” supervised-labelling function (colour face crops)

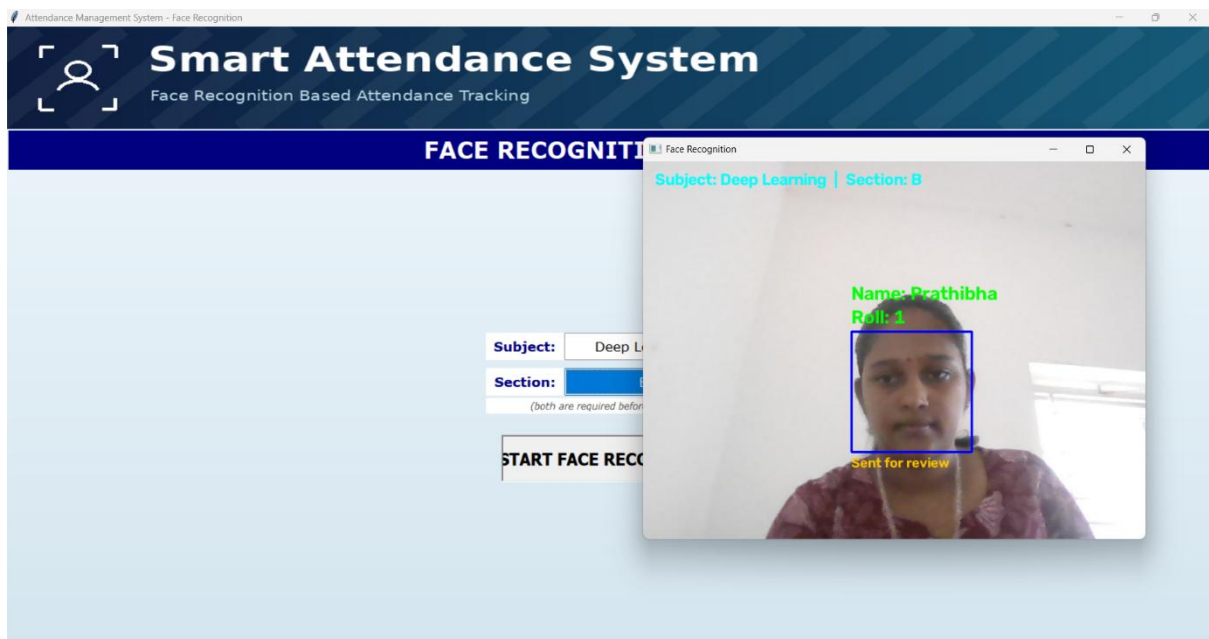


Figure 5.4: Live face recognition window, showing CNN-based recognition with subject/section entry and bounding-box overlay

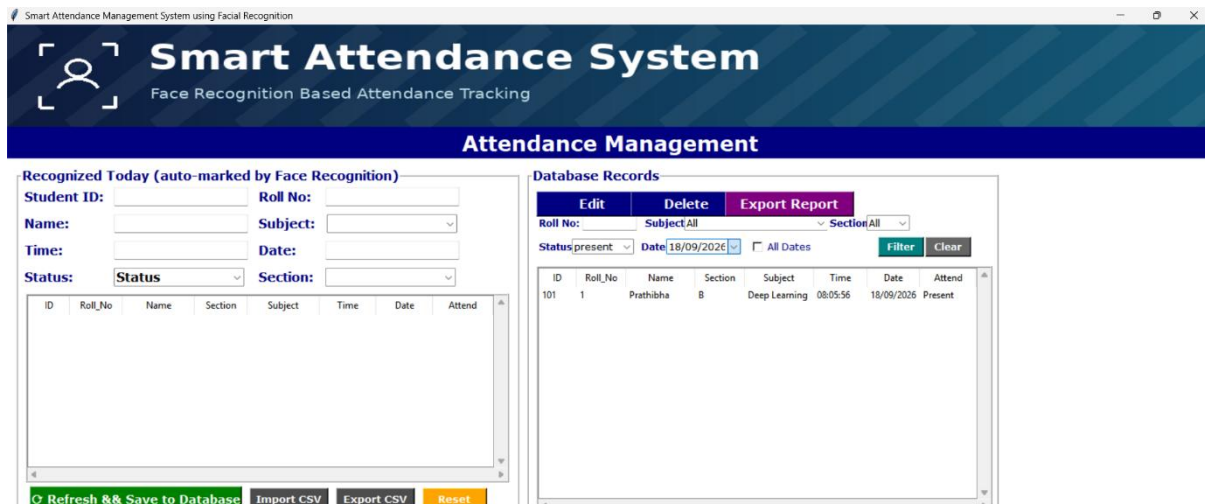


Figure 5.5: Attendance management screen — recognised queue, database records, and export

5.6 Societal Impact of the Project

Automating attendance returns instructional time that would otherwise be spent on roll calls or circulating a signature sheet, and removes a routine opportunity for proxy attendance, an integrity issue every manual attendance method shares to some degree. Beyond time savings, a structured, queryable attendance record makes it straightforward for a department to identify students whose attendance is trending downward across a semester, which is often an early, actionable signal of a student needing academic or personal support — a use of the same underlying database that goes beyond the immediate goal of automating a single class’s roll call.

The project also carries societal responsibilities that a purely technical evaluation can understate. Facial recognition is biometric data, and this project currently stores captured face images and derived embeddings as plain files on the deployment machine rather than encrypted at rest, which is an acceptable posture for a lab prototype but not for a production deployment handling real students’ biometric data without explicit consent and retention policies — addressed further as a limitation in Section 6. It is also worth acknowledging a known, general limitation of face detection and recognition systems: published research has documented that face detectors and recognisers, including ones built on Haar Cascades and CNN embeddings alike, can exhibit uneven accuracy across skin tones and lighting conditions if their underlying training or calibration data is not representative. This project has not specifically measured

such disparities in its own deployment, and doing so — rather than assuming the system is unbiased — is listed as future work in Section 6.

5.7 SDG Mapped

This project maps most directly to United Nations Sustainable Development Goal 4 (Quality Education), by reducing the administrative time that automated attendance takes away from teaching, and by enabling the kind of attendance-trend analytics described in Section 5.6 that can support earlier intervention for at-risk students. It also relates to SDG 9 (Industry, Innovation and Infrastructure), in that the project applies a deep learning technique — CNN-based face embeddings — to build a small piece of institutional digital infrastructure, and documents, as a case study, how such a system’s recognition component can be upgraded from a classical to a deep-learning method without discarding the surrounding infrastructure already built. A weaker, secondary connection can be drawn to SDG 8 (Decent Work and Economic Growth), insofar as automating a repetitive administrative task is a small instance of the broader productivity gains associated with appropriate automation; this connection is included for completeness rather than claimed as a central contribution of the project.

SDG	Relevance
SDG 4 — Quality Education	Reduces time spent on administrative attendance-taking; enables attendance-trend analytics that can support earlier academic intervention.
SDG 9 — Industry, Innovation and Infrastructure	Applies CNN-based deep learning to build reusable institutional infrastructure, with a documented, low-disruption path to upgrading the recognition method itself.
SDG 8 — Decent Work and Economic Growth	Minor, secondary relevance: automates a repetitive administrative task, freeing staff and faculty time for higher-value work.

Table 5.4: SDGs mapped to this project

6. Conclusion and Future Scope

This project delivered a complete, functioning Smart Attendance System that automates classroom attendance using deep learning-based facial recognition, spanning student enrolment, supervised dataset creation, CNN-based embedding extraction, real-time recognition, database synchronisation with administrator oversight, and exportable reporting. The recognition engine, built on a Haar Cascade detector and a pretrained FaceNet CNN accessed via the deepface library, replaces an earlier classical LBPH-based version of this system; the migration was carried out without altering any of the surrounding modules, demonstrating the practical value of the project's original decision to isolate the recognition engine behind a small, well-defined interface (Section 4.2).

Future Scope

The following improvements can be considered in future development of the Smart Attendance System:

- 1. ArcFace-based Recognition:** Replace FaceNet with ArcFace to obtain more discriminative facial embeddings and improve recognition under challenging conditions.
- 2. Advanced Face Detection:** Replace the Haar Cascade detector with a CNN-based detector such as MTCNN or RetinaFace for more robust face detection.
- 3. Liveness Detection:** Integrate liveness or anti-spoofing techniques to prevent unauthorized attendance using printed photographs or other presentation attacks.
- 4. Incremental Student Enrollment:** Allow new facial samples to be added for individual students without requiring the complete embedding dataset to be recomputed.
- 5. Real-Time Accuracy Evaluation:** Conduct a formal live evaluation of the CNN-based recognition system under different classroom conditions to measure accuracy, precision, recall, and F1-score.

- 6. Biometric Data Privacy:** Implement secure storage of facial embeddings, encryption, user consent, and data-retention policies to improve the privacy and security of biometric information.
- 7. Robustness and Fairness Evaluation:** Evaluate recognition performance under different lighting conditions, facial poses, camera qualities, and demographic variations.
- 8. Multi-Camera Support:** Extend the system to support multiple cameras for larger classrooms and simultaneous attendance monitoring.
- 9. Mobile Application:** Develop a mobile companion application that allows administrators to monitor attendance, manage students, and receive notifications remotely.
- 10. Attendance Analytics:** Add dashboards for analysing attendance trends by student, subject, and semester, with graphical reports and automated summaries.

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